

Mean and Variance in Time Series

To understand **Stationarity**, you must know **Mean** and **Variance**.

1) Mean (Average)

Definition

The **Mean** is the average value of all observations.

Formula

Formula

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

Where:

- x = observations
 - n = number of observations
-

Example

Daily sales:

Day	Sales
Mon	100
Tue	120
Wed	140
Thu	160
Fri	180

Calculation

$$\text{Mean} = (100 + 120 + 140 + 160 + 180) / 5$$

$$\text{Mean} = 700 / 5$$

$$\text{Mean} = 140$$

✓ Average sales = **140**

Real-Time Example of Mean

Suppose a student gets:

Subject	Mark
---------	------

Maths	80
-------	----

Science	90
---------	----

English	70
---------	----

$$\text{Mean} = (80 + 90 + 70) / 3$$

$$\text{Mean} = 240 / 3 = 80$$

✓ Average mark = **80**

2) Variance

Definition

Variance measures how far the data points are spread from the mean.

- Small Variance → Data points are close to mean.
- Large Variance → Data points are far from mean.

Formula

Formula

$$\text{Variance} = \frac{\sum (x_i - \bar{x})^2}{n}$$

Example 1 (Low Variance)

Data:

98, 100, 102

Step 1: Mean

$$(98+100+102)/3 = 100$$

Step 2: Difference from Mean

Value	Mean	Difference
98	100	-2
100	100	0
102	100	+2

Step 3: Square Differences

Difference	Square
-2	4
0	0
2	4

$$\text{Variance} = (4+0+4)/3$$

$$\text{Variance} = 2.67$$

✓ Small variance

The values are very close together.

Example 2 (High Variance)

Data:

20, 100, 180

Step 1: Mean

$$(20+100+180)/3 = 100$$

Step 2

Value	Mean	Difference
20	100	-80
100	100	0
180	100	+80

Step 3

Difference	Square
-80	6400
0	0
80	6400

$$\text{Variance} = (6400+0+6400)/3$$

$$\text{Variance} = 4266.67$$

✗ Very large variance

Values are spread far apart.

Why Mean and Variance Matter in Time Series?

Stationary Series

Year	Sales
1	100
2	102
3	98

4 101

5 99

Mean $\approx$ 100

Variance $\approx$ Constant

 Stationary

Non-Stationary Series

Year	Sales
------	-------

1	100
---	-----

2	200
---	-----

3	300
---	-----

4	400
---	-----

5	500
---	-----

Mean keeps changing over time.

 Non-Stationary

Easy Real-Life Analogy

Mean = Average Salary

Suppose 5 employees earn:

25000, 26000, 25500, 24500, 25000

Mean salary $\approx$ ₹25,200

Variance = Salary Difference

If salaries are:

25000, 26000, 25500, 24500, 25000

Variance is small 

If salaries are:

10000, 25000, 50000, 75000, 100000

Variance is huge 

Mean: Average value of the observations.

Variance: Measure of how much the observations vary from the mean.

For a **stationary time series**, both the **mean and variance should remain approximately constant over time**.

=====